

Questions with Answer Keys

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Q1: 24 Feb (Shift 1) - Single Correct

$\lim_{x \rightarrow 0} \frac{\int_0^{x^2} (\sin \sqrt{t}) dt}{x^3}$ is equal to:

(1) $\frac{2}{3}$

(2) 0

(3) $\frac{1}{15}$

(4) $\frac{3}{2}$

Q2: 24 Feb (Shift 1) - Numerical

If $\int_{-a}^a (|x| + |x - 2|) dx = 22$, ($a > 2$) and $[x]$ denotes the greatest integer $\leq x$, then $\int_a^{-a} (x + [x]) dx$ is equal to

Q3: 24 Feb (Shift 2) - Single Correct

The value of the integral, $\int_1^3 [x^2 - 2x - 2] dx$, where $[x]$ denotes the greatest integer less than or equal to x , is:

(1) -4

(2) -5

(3) $-\sqrt{2} - \sqrt{3} - 1$

(4) $-\sqrt{2} - \sqrt{3} + 1$

Q4: 24 Feb (Shift 2) - Single Correct

Let $f(x)$ be a differentiable function defined on $[0, 2]$ such that $f'(x) = f'(2 - x)$ for all $x \in (0, 2)$, $f(0) = 1$ and $f(2) = e^2$. Then the value of $\int_0^2 f(x) dx$ is:

(1) $1 + e^2$

(2) $1 - e^2$

(3) $2(1 - e^2)$

(4) $2(1 + e^2)$

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Q5: 25 Feb (Shift 1) - Single Correct

The value of $\int_{-1}^1 x^2 e^{[x^3]} dx$, where $[t]$ denotes the greatest integer $\leq t$, is :

- (1) $\frac{e+1}{3}$
- (2) $\frac{e-1}{3e}$
- (3) $\frac{e+1}{3e}$
- (4) $\frac{1}{3e}$

Q6: 25 Feb (Shift 2) - Single Correct

If $I_n = \int_{\pi/4}^{\pi/2} \cot^n x dx$, then:

- (1) $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}$ are in G.P.
- (2) $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}$ are in A.P.
- (3) $I_2 + I_4, I_3 + I_5, I_4 + I_6$ are in A.P.
- (4) $I_2 + I_4, (I_3 + I_5)^2, I_4 + I_6$ are in G.P.

Q7: 25 Feb (Shift 2) - Single Correct

$\lim_{x \rightarrow \infty} \left[\frac{1}{n} + \frac{n}{(n+1)^2} + \frac{n}{(n+2)^2} + \dots + \frac{n}{(2n-1)^2} \right]$ is equal to:

- (1) 1
- (2) $\frac{1}{3}$
- (3) $\frac{1}{2}$
- (4) $\frac{1}{4}$

Q8: 25 Feb (Shift 2) - Numerical

The value of $\int_2^3 |3x^2 - 3x - 6| dx$ is

Q9: 26 Feb (Shift 1) - Single Correct

The value of $\sum_{n=1}^{100} \int_{n-1}^n e^{x-[x]} dx$, where $[x]$ is the greatest integer $\leq x$, is:

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(1) $100(e - 1)$

(2) $100e$

(3) $100(1 - e)$

(4) $100(1 + e)$

Q10: 26 Feb (Shift 1) - Single Correct

The value of $\int_{-\pi/2}^{\pi/2} \frac{\cos^2 x}{1+3^x} dx$ is:

(1) 2π

(2) 4π

(3) $\frac{\pi}{2}$

(4) $\frac{\pi}{4}$

Q11: 26 Feb (Shift 1) - Numerical

The value of the integral $\int_0^\pi |\sin 2x| dx$ is

Q12: 26 Feb (Shift 2) - Single Correct

For $x > 0$, if $f(x) = \int_1^x \frac{\log_e t}{(1+t)} dt$, then $f(e) + f\left(\frac{1}{e}\right)$ is equal to :

(1) $\frac{1}{2}$

(2) -1

(3) 1

(4) 0

Q13: 26 Feb (Shift 2) - Single Correct

Let $f(x) = \int_0^x e^t f(t) dt + e^x$ be a differentiable function for all $x \in R$. Then $f(x)$ equals.

(1) $2e^{(e^x-1)} - 1$

(2) $e^{(e^x-1)}$

(3) $2e^{e^x} - 1$

$$(4) e^{e^x} - 1$$

Q14: 26 Feb (Shift 2) - Numerical

In $I_{m,n} = \int_0^1 x^{m-1}(1-x)^{n-1}dx$, for $m, n \geq 1$ and $\int_0^1 \frac{x^{m-1}+x^{n-1}}{(1+x)^{m+n}}dx = \alpha I_{m,n}$, $\alpha \in \mathbb{R}$, then α equals

Answer Key**Q1 (1)****Q2 (3)****Q3 (3)****Q4 (1)****Q5 (3)****Q6 (2)****Q7 (3)****Q8 (19)****Q9 (1)****Q10 (4)****Q11 (2)****Q12 (1)****Q13 (1)****Q14 (1)**

Q1 (1)

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\int_0^{x^2} \sin \sqrt{t} dt}{x^3} \\ = \lim_{x \rightarrow 0} \frac{(\sin |x|) 2x}{3x^2} \\ = \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \times \frac{2}{3} \\ = \frac{2}{3} \end{aligned}$$

Q2 (3)

$$\int_{-a}^0 (-2x + 2) dx + \int_0^2 (x + 2 - x) dx + \int_2^a (2x - 2) dx = 22$$

$$x^2 - 2x \Big|_0^{-a} + 2x \Big|_0^2 + x^2 - 2x \Big|_2^a = 22$$

$$a^2 + 2a + 4 + a^2 - 2a - (4 - 4) = 22$$

$$2a^2 = 18 \Rightarrow a = 3$$

$$\begin{aligned} \int_3^{-3} (x + [x]) dx &= - \left(\int_{-3}^3 (x + [x]) dx \right) = - \left(\int_{-3}^3 [x] dx \right) \\ &= -(-3 - 2 - 1 + 0 + 1 + 2) = 3 \end{aligned}$$

Q3 (3)

$$I = \int_1^3 -3dx + \int_1^3 [(x-1)^2] dx$$

$$\text{Put } x-1 = t; dx = dt$$

$$I = (-6) + \int_0^2 [t^2] dt$$

$$I = -6 + \int_0^1 0dt + \int_1^{\sqrt{2}} 1dt + \int_{\sqrt{2}}^{\sqrt{3}} 2dt + \int_{\sqrt{3}}^2 3dt$$

$$I = -6 + (\sqrt{2} - 1) + 2\sqrt{3} - 2\sqrt{2} + 6 - 3\sqrt{3}$$

$$I = -1 - \sqrt{2} - \sqrt{3}$$

Q4 (1)

$$f'(x) = f'(2-x)$$

$$\text{On integrating both side } f(x) = -f(2-x) + c$$

$$\text{put } x = 0 \quad f(0) + f(2) = c \Rightarrow c = 1 + e^2$$

$$\Rightarrow f(x) + f(2-x) = 1 + e^2 \dots \dots (i)$$

$$I = \int_0^2 f(x) dx = \int_0^1 \{f(x) + f(2-x)\} dx = (1 + e^2)$$

Q5 (3)

$$I = \int_{-1}^0 x^2 \cdot e^{-1} dx + \int_0^1 x^2 dx$$

$$\therefore I = \frac{x^3}{3e} \Big|_{-1}^0 + \frac{x^3}{3} \Big|_0^1$$

$$\Rightarrow I = \frac{1}{3e} + \frac{1}{3}$$

Q6 (2)

$$I_{n+2} + I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cot^n x \cdot \operatorname{cosec}^2 x dx = \left[\frac{-(\cot x)^{n+1}}{n+1} \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$I_{n+2} + I_n = \frac{1}{n+1}$$

$$I_2 + I_4 = \frac{1}{3}, I_3 + I_5 = \frac{1}{4}, I_4 + I_6 = \frac{1}{5}$$

Q7 (3)

$$\lim_{x \rightarrow \infty} \sum_{r=0}^{n-1} \frac{n}{(n+r)^2} = \lim_{x \rightarrow \infty} \sum_{r=0}^{n-1} \frac{n^2}{n^2(1+\frac{r}{n})^2} = \int_0^1 \frac{dx}{(1+x)^2}$$

$$= -\left[\frac{1}{1+x} \right]_0^1 \Rightarrow -\left[\frac{1}{2} - 1 \right] = \frac{1}{2}$$

Q8 (19)

$$3 \int_{-2}^2 |x^2 - x - 2| dx \quad x^2 - x - 2 = (x-2)(x+1)$$



$$= 3 \left\{ \int_{-2}^{-1} (x^2 - x - 2) dx + \int_{-1}^2 (-x^2 + x + 2) dx \right\}$$

$$= 3 \left[\left(\frac{x^3}{3} - \frac{x^2}{2} - 2x \right)_{-2}^{-1} - \left(\frac{x^3}{3} - \frac{x^2}{2} - 2x \right)_{-1}^2 \right]$$

$$= 19$$

Q9 (1)

$$\sum_{n=1}^{100} \int_{n-1}^n e^{x-[x]} dx$$

$$= \int_0^1 e^{\{x\}} dx + \int_1^2 e^{\{x\}} dx + \int_2^3 e^{\{x\}} dx + \dots + \int_{99}^{100} e^{\{x\}} dx \quad (\because \{x\} = x - [x])$$

Hints and Solutions

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$$= e^x|_0^1 + e^{(x-1)}|_1^2 + e^{(x-2)}|_2^3 + \dots + e^{(x-99)}|_{99}^{100}$$

$$= (e-1) + (e-1) + (e-1) + \dots + (e-1)$$

$$= 100(e-1)$$

Q10 (4)

Let $I = \int_{-\pi/2}^{\pi/2} \frac{\cos^2 x}{1+3^x} dx$

$$I = \int_{-\pi/2}^{\pi/2} \frac{3^x \cos^2 x}{1+3^x} dx$$

$$2I = \int_{-\pi/2}^{\pi/2} \cos^2 x dx$$

$$I = \int_0^{\pi/2} \cos^2 x dx = \frac{\pi}{4}$$

Q11 (2)

$$I = \int_0^{\pi} |\sin 2x| dx$$

$$I = 2 \int_0^{\pi/2} |\sin 2x| dx = 2 \int_0^{\pi/2} \sin 2x dx$$

$$I = 2 \left[\frac{-\cos(2x)}{2} \right]_0^{\pi/2} = 2$$

Q12 (1)

$$f(e) + f\left(\frac{1}{e}\right) = \int_1^e \frac{\ln t}{1+t} dt + \int_1^{1/e} \frac{\ln t}{1+t} dt = I_1 + I_2$$

$$I_2 = \int_1^{1/e} \frac{\ln t}{1+t} dt \quad \text{put } t = \frac{1}{z}, dt = -\frac{dz}{z^2}$$

$$= \int_1^e -\frac{\ln z}{1+\frac{1}{z}} \times \left(-\frac{dz}{z^2}\right) = \int_1^e \frac{\ln z}{z(z+1)} dz$$

$$f(e) + f\left(\frac{1}{e}\right) = \int_1^e \frac{\ln t}{1+t} dt + \int_1^e \frac{\ln t}{t(t+1)} dt = \int_1^e \frac{\ln t}{1+t} + \frac{\ln t}{t(t+1)} dt$$

$$= \int_1^e \frac{\ln t}{t} dt \left\{ \ln t = u, \frac{1}{t} dt \right\}$$

$$= du = \int_0^1 u du = \frac{u^2}{2} \Big|_0^1 = \frac{1}{2}$$

Q13 (1)

Given, $f(x) = \int_0^x e^t f(t) dt + e^x \dots (1)$

Differentiating both sides w.r.t x $f'(x) = e^x \cdot f(x) + e^x$

(Using Newton Leibnitz Theorem) $\Rightarrow \frac{f'(x)}{f(x)+1} = e^x$

Integrating w.r.t x $\int \frac{f'(x)}{f(x)+1} dx = \int e^x dx$

$$\Rightarrow \ln(f(x) + 1) = e^x + c$$

Put $x = 0$

$$2 = 1 + c \quad (\because f(0) = 1, \text{ from equation (1)})$$

$$\therefore \ln(f(x) + 1) = e^x + \ln 2 - 1$$

$$\Rightarrow f(x) + 1 = 2 \cdot e^{e^x - 1}$$

$$\Rightarrow f(x) = 2e^{e^x - 1} - 1$$

Q14 (1)

$$I_{m,n} = \int_0^1 x^{m-1} \cdot (1-x)^{n-1} dx$$

$$\text{Put } x = \frac{y}{y+1} \Rightarrow dx = \frac{-1}{(y+1)^2} dy$$

$$1-x = \frac{y}{y+1}$$

$$\therefore I_{m,n} = \int_\infty^0 \frac{y^{m-1}}{(y+1)^{m+n}} (-1) dy = \int_0^\infty \frac{y^{m-1}}{(y+1)^{m+n}} dy \dots (i)$$

$$\text{Similarly } I_{m,n} = \int_0^1 x^{n-1} \cdot (1-x)^{m-1} dx$$

$$\Rightarrow I_{m,n} = \int_0^\infty \frac{y^{n-1}}{(y+1)^{m+n}} dy \dots (ii)$$

From (i) & (ii)

$$2I_{m,n} = \int_0^\infty \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy$$

$$\Rightarrow 2I_{m,n} = \int_0^1 \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy + \int_1^\infty \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy$$

$$\text{Put } y = \frac{1}{z} \text{ in } I_2$$

$$dy = -\frac{1}{z^2} dz$$

$$\Rightarrow 2I_{m,n} = \int_0^1 \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy + \int_1^0 \frac{z^{m-1} + z^{n-1}}{(z+1)^{m+n}} (-dz)$$

$$\Rightarrow I_{m,n} = \int_0^1 \frac{y^{m-1} + y^{n-1}}{(y+1)^{m+n}} dy \Rightarrow \alpha = 1$$